Total number of printed pages – 4

23R–25 (CSE3204C)

**2025**

**~~MATHEMATICS~~** CSE

(MAJOR)

Paper : CSE3204C

***( Discrete Mathematics )***

*Full Marks : 60*

Time : 2½ hours

***The figures in the margin indicate full marks for the questions.***

1. Answer the following questions : 1×7=7

(a) What is subset ?

(b) Define domain of a relation.

(c) What is function ?

(d) Define subgraph.

(e) What is group ?

(f) Define tree.

(g) What is combination ?

2. Answer the following questions : 2×4=8

(a) How many types of sets are there? Define each.

(b) Define complete graph with example.

(c) What are the basic rules of counting?

(d) How many permutations are there of the letter 'DISTINCT'?

3. Answer the following questions :

(a) Define bipartite and complete bipartite graph with examples. 5

(b) Show that the relation $R$ in the set $Z$ of integers given by $R = \{(a,b) : 2 \text{ divides } a-b\}$ is an equivalence relation. 5

**Or**

(c) How many types of functions are there? Define each with example.

(d) What is isomorphic graph? Give example. 5

**Or**

(e) Define isolated point and pendent vertex with suitable example. 5

4. Answer the following questions: ***(any one)*** $10 \times 1 = 10$

*(a)* A man has 7 relatives, 4 of them are ladies and 3 gentlemen, his wife has 7 relatives, 3 of them are ladies and 4 gentlemen. In how many ways can they invite a dinner party of 3 ladies and 3 gentlemen so that there are 3 of man's relatives and 3 of wife's relatives ?

*(b)* Show that the set

$$G = \{----2^{-4}, 2^{-3}, 2^{-2}, 2^{-1}, 1, 2, 2^{2}, 2^{3}, ----\}$$

is a group w.r.t. multiplication.

5. Answer the following questions: ***(any one)*** $10 \times 1 = 10$

*(a)* If $X$ and $Y$ are *two* sets such that $X \cup Y$ has 50 elements $X$ has 28 elements and $Y$ has 32 elements, how many elements does $X \cap Y$ have ? use Venn diagram.

*(b)* Prove that a graph $G$ is a tree iff there is one and only one unique path between *any two* vertices.

6. Answer **any one** of the following questions: $10 \times 1 = 10$

(a) Let $P(X)$ be the power of a non empty set X. Examine if $P(X)$ is a group under the composition $*$ defined by $A * B = (A - B) \cup (B - A), A, B, \in, P(X)$.

(b) Give an example of a relation, which is

(i) Symmetric but neither reflexive nor transitive.

(ii) Reflexive and transitive but not symmetric.